

Library Platform — Architecture & Cost Brief

Public book download & read · target ~100,000 visitors/minute · catalogue under ~10 GB · Prepared Aug 2026

Objective. A production-grade, SEO-friendly library website where any user can browse, read, and download books, engineered to absorb a ~100k-visitors-per-minute launch spike, with a catalogue kept under ~10 GB total.

CORE PRINCIPLE

The 100k/min target is a **caching problem, not a server-scaling problem**. A book catalogue is read-heavy and rarely changes, so pages are pre-rendered and served from Cloudflare's edge, and files are delivered from R2 with zero egress fees. Handled this way, the spike costs roughly the same as light traffic.

RECOMMENDED STACK

LAYER	CHOICE	ROLE
Domain · CDN · WAF · cache	Cloudflare (DNS on Cloudflare)	Absorbs the spike, caches pages & files at edge
Book files (download + read)	Cloudflare R2	\$0 egress at any volume; each file cached at edge
App & pages	Next.js on Cloudflare Pages	SSG/ISR catalogue + small dynamic APIs
Catalogue database	Postgres (managed or self-hosted)	Owned, portable, extensible; holds metadata + admin
Reader / download	PDF.js (range requests) + download Worker	Stream to read in-browser; one-click download

KEY DESIGN RULES

- Books never touch a build.** Admin uploads a book → file goes to R2 + a metadata row to Postgres → the page revalidates via on-demand ISR. **No git commit, no app rebuild.**
- Code and content are separate paths.** Git push builds the app (rare); book management is a runtime admin flow (frequent). GitHub Actions is used only for app deploys and an optional one-time bulk-ingest of the initial catalogue to R2.
- ISR keeps growth free.** Any catalogue size builds the same way; adding a title never triggers a full rebuild.
- Custom domain is free** — the tier is not the blocker; the domain's nameservers simply need to point at Cloudflare.

CATALOGUE SIZE (UNDER ~10 GB)

Kept under ~10 GB, storage sits **entirely inside R2's free tier** — so storage cost is **\$0**. Egress stays \$0, and edge-cached pages serve identically regardless of catalogue size. Note the 10 GB free allowance is storage-at-rest: higher-res covers, page-image tiles for the reader, or extra formats (PDF + EPUB) can quietly add up, and anything past 10 GB is only \$0.015/GB.

MONTHLY COST SUMMARY

NEED	PLAN	COST
Custom domain	Cloudflare Free (DNS on Cloudflare)	\$0
App builds (code only)	Pages Free — 500 builds/mo	\$0
Static page serving	Pages Free — unlimited	\$0
Book files (under ~10 GB)	R2 — within 10 GB free tier, \$0 egress	\$0
Dynamic API · admin · downloads at spike	Workers Paid — 10M req + 30M CPU-ms	\$5
<i>Optional:</i> WAF · analytics · rate limiting	Pro zone plan	\$0-20

Bare minimum: ~\$5 / month Comfortable production: ~\$5-25 / month

 **Credit-card verification required**

To activate the paid pieces above, Cloudflare requires a valid **credit/debit card added and verified** on the account. This is needed to enable **Workers Paid (\$5/mo)** and to switch on **R2** (which stays within the free 10 GB here, so \$0 storage). The card must be on file before these services can be turned on — even though actual usage cost is only ~\$5/month. No card is needed for the free tiers alone (custom domain, static hosting, 500 builds/mo).

NEXT STEPS

- Move the domain's DNS to Cloudflare (free) and attach it to the Pages project.
- Add & verify a payment method → enable Workers Paid + R2.
- Build the admin upload flow (presigned R2 upload → Postgres row → on-demand ISR revalidate).
- Bulk-ingest the initial catalogue (under ~10 GB) to R2 via a one-time GitHub Action / wrangler r2.
- Add Turnstile + rate-limiting on the download endpoint; load-test to confirm 100k/min.